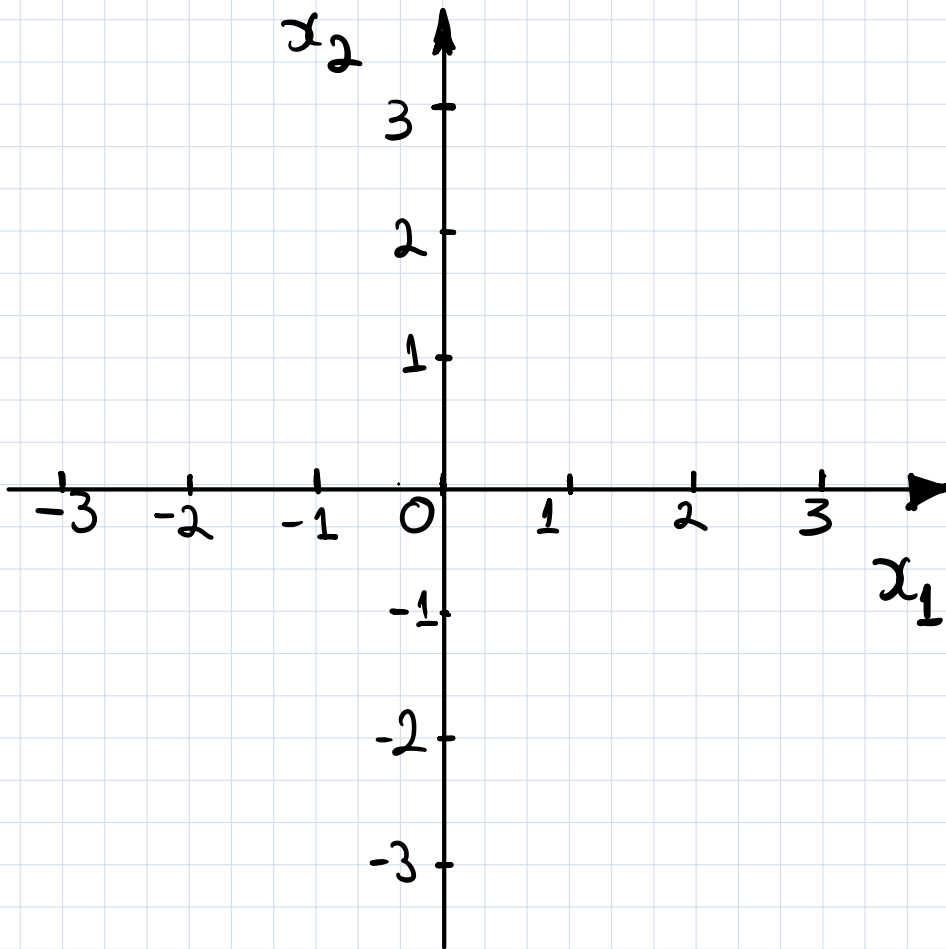


Plot points



Plot:

$(0,0)$

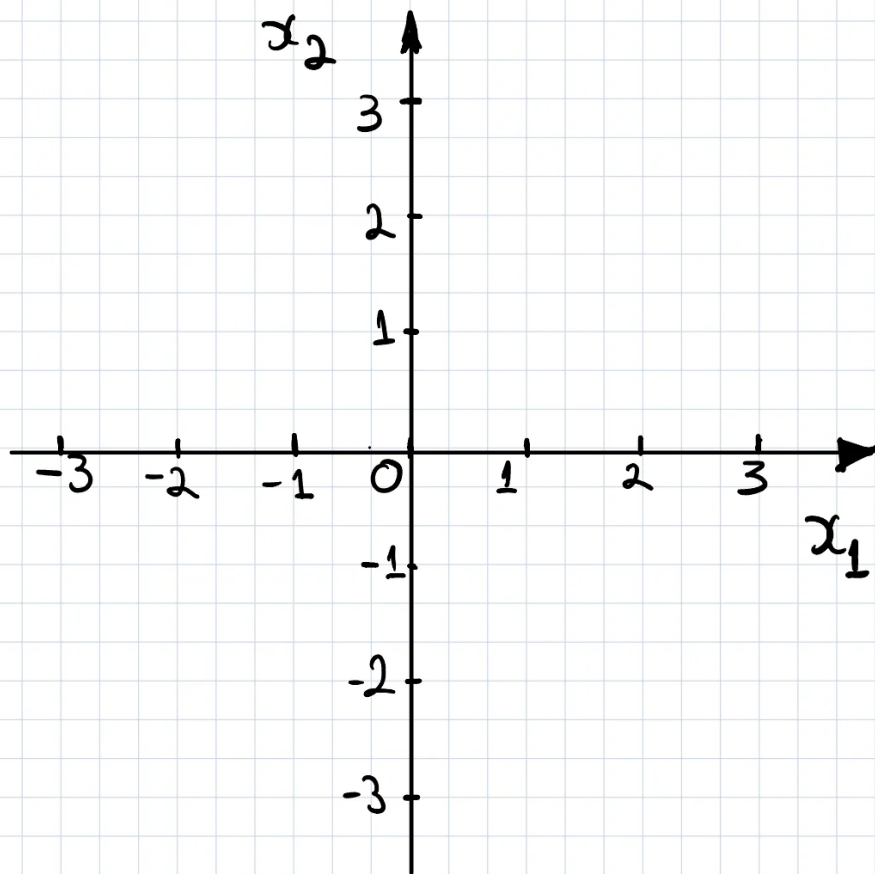
$(1,2)$

$(3,4)$

$(-1,2)$

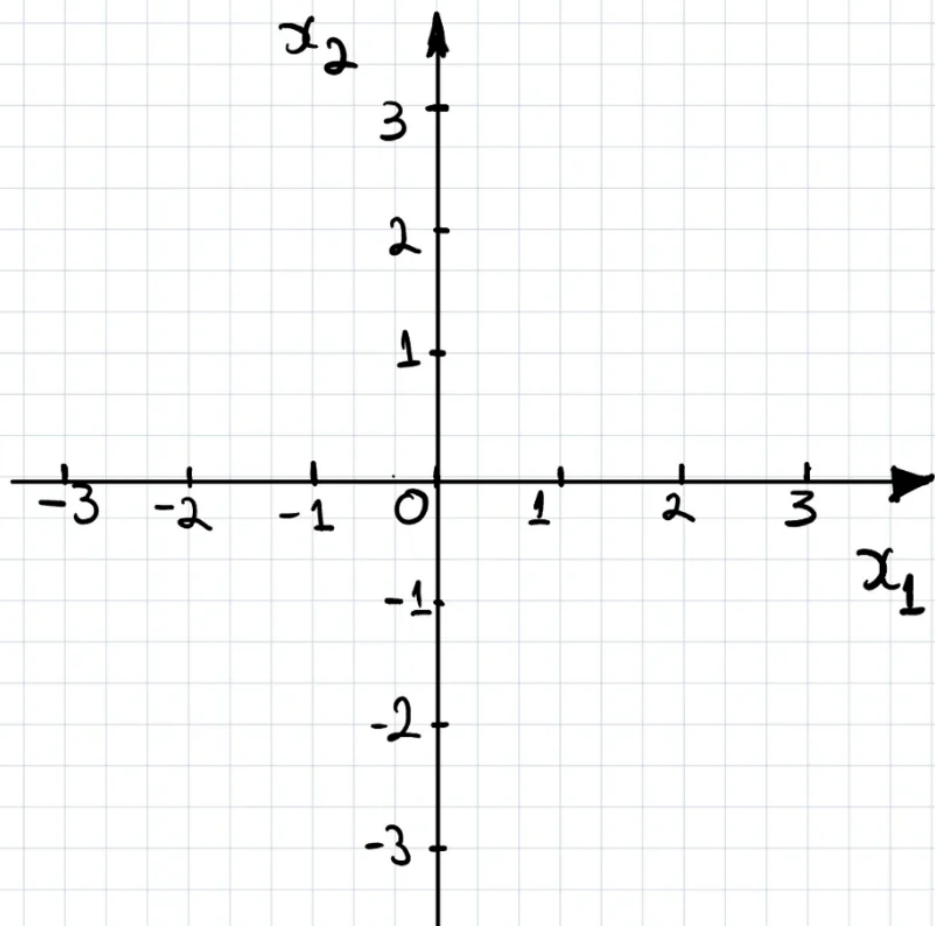
$(-3,-1)$

Inequalities

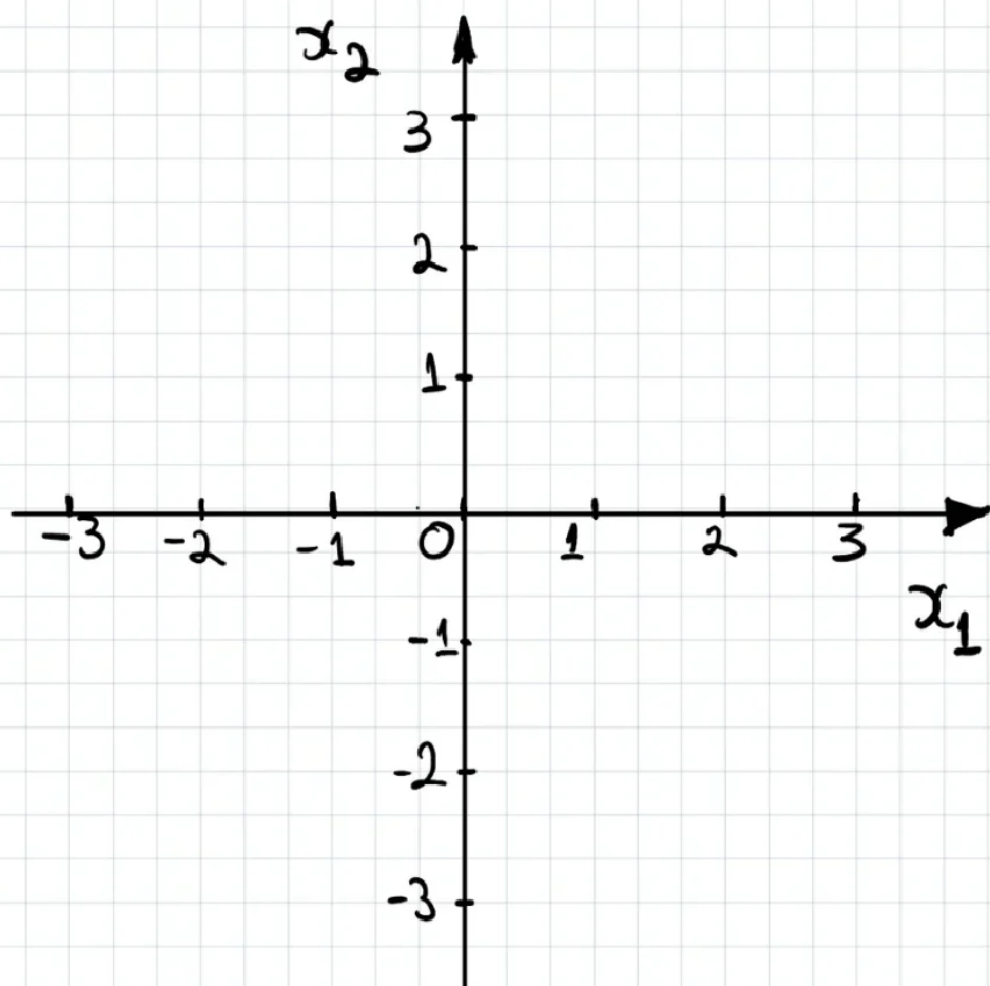


Plot:

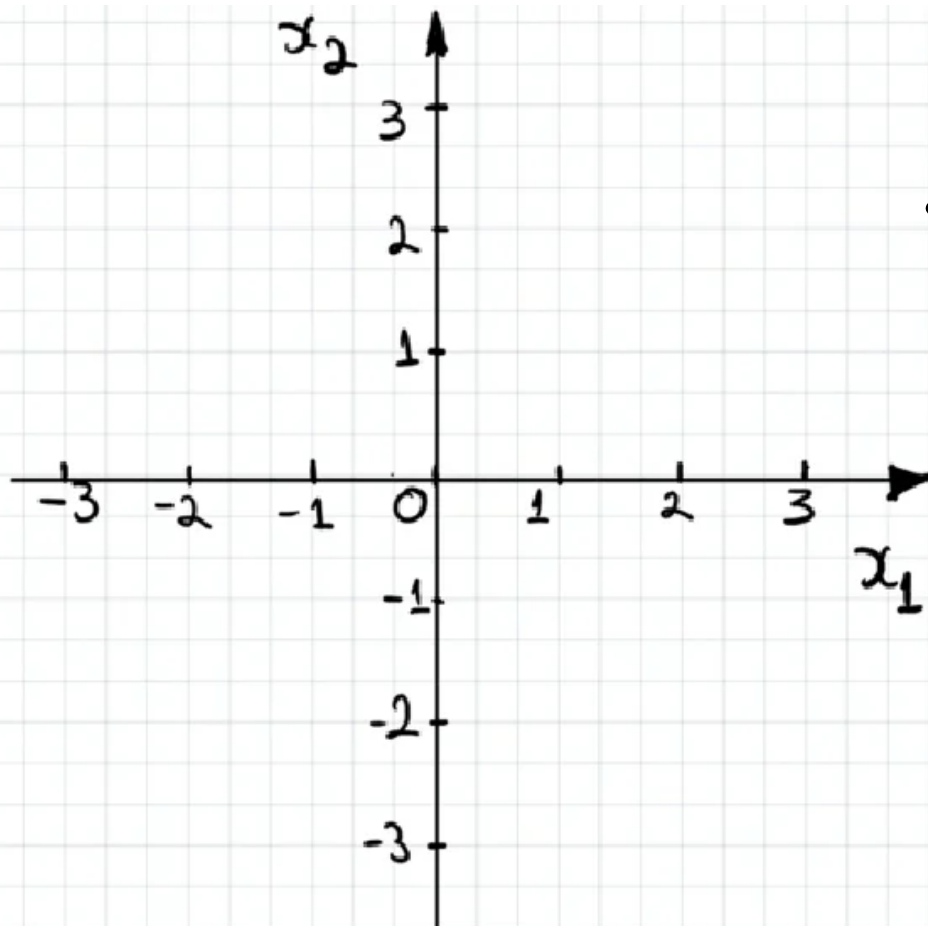
$$x_1 \geq 0$$



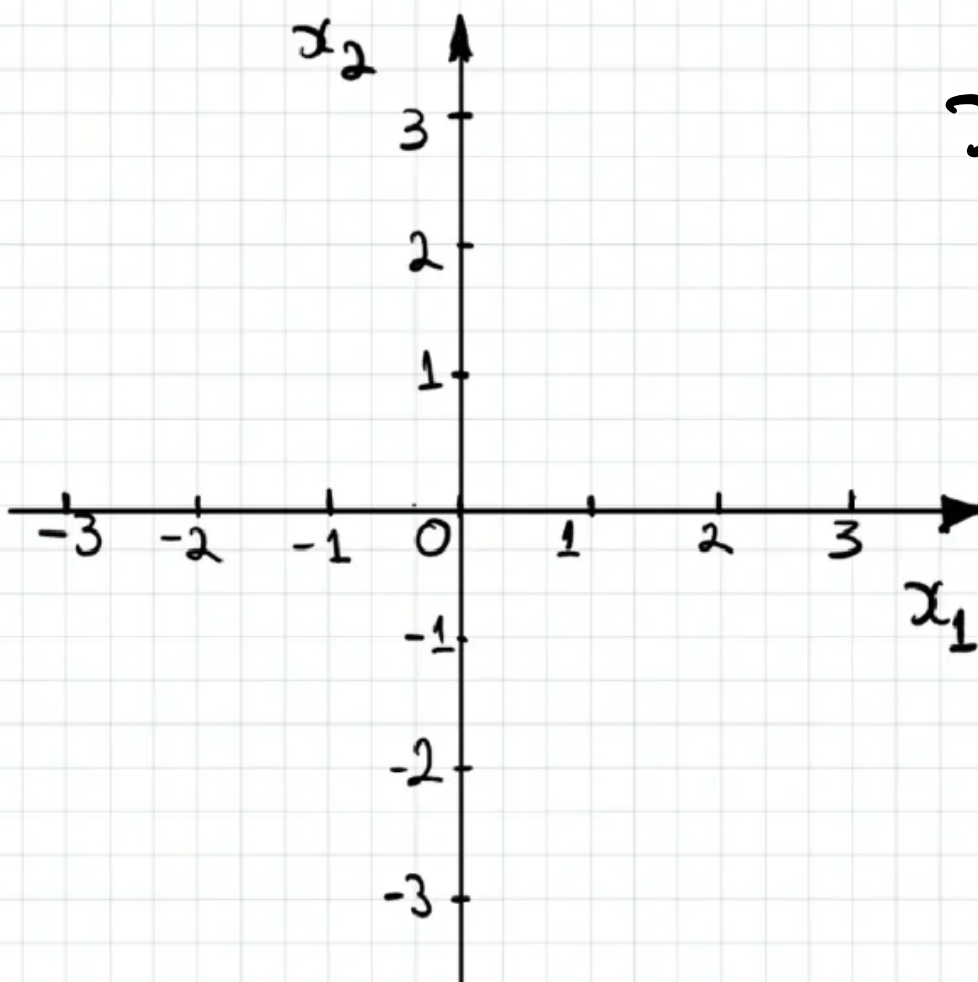
$$x_2 \geq 0$$



$$x_2 \geq x_1$$



$$x_2 \geq x_1 + 1$$



$$x_2 \geq 2 - x_1$$

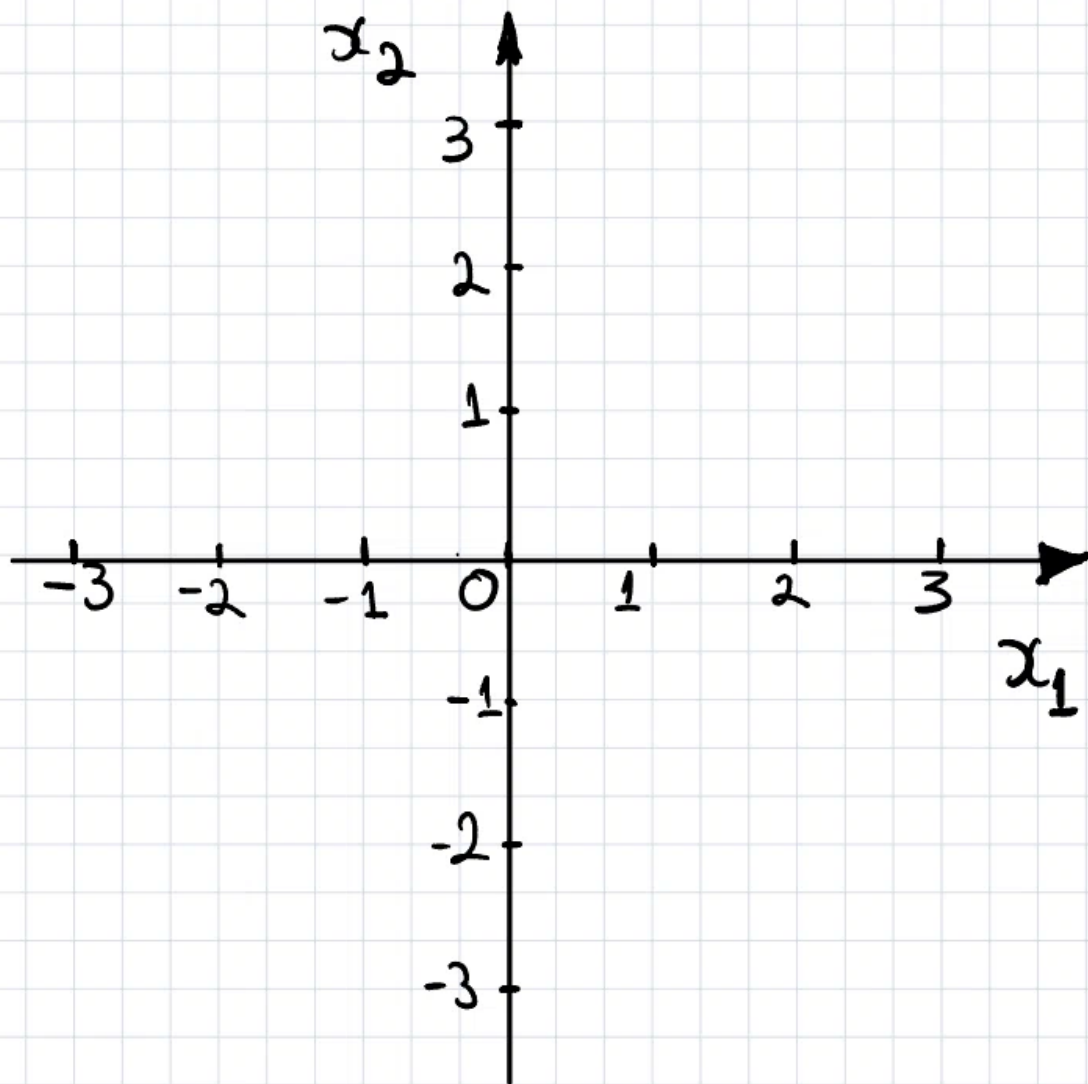
Multiple inequalities make regions.

$$x_1 \geq 0$$

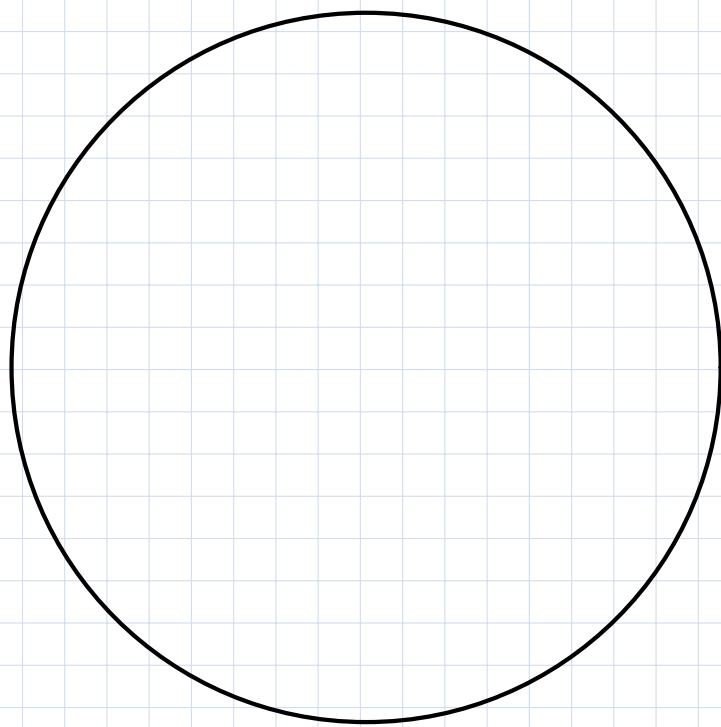
$$x_2 \geq 0$$

$$x_1 \leq 1$$

$$x_2 \leq 1$$

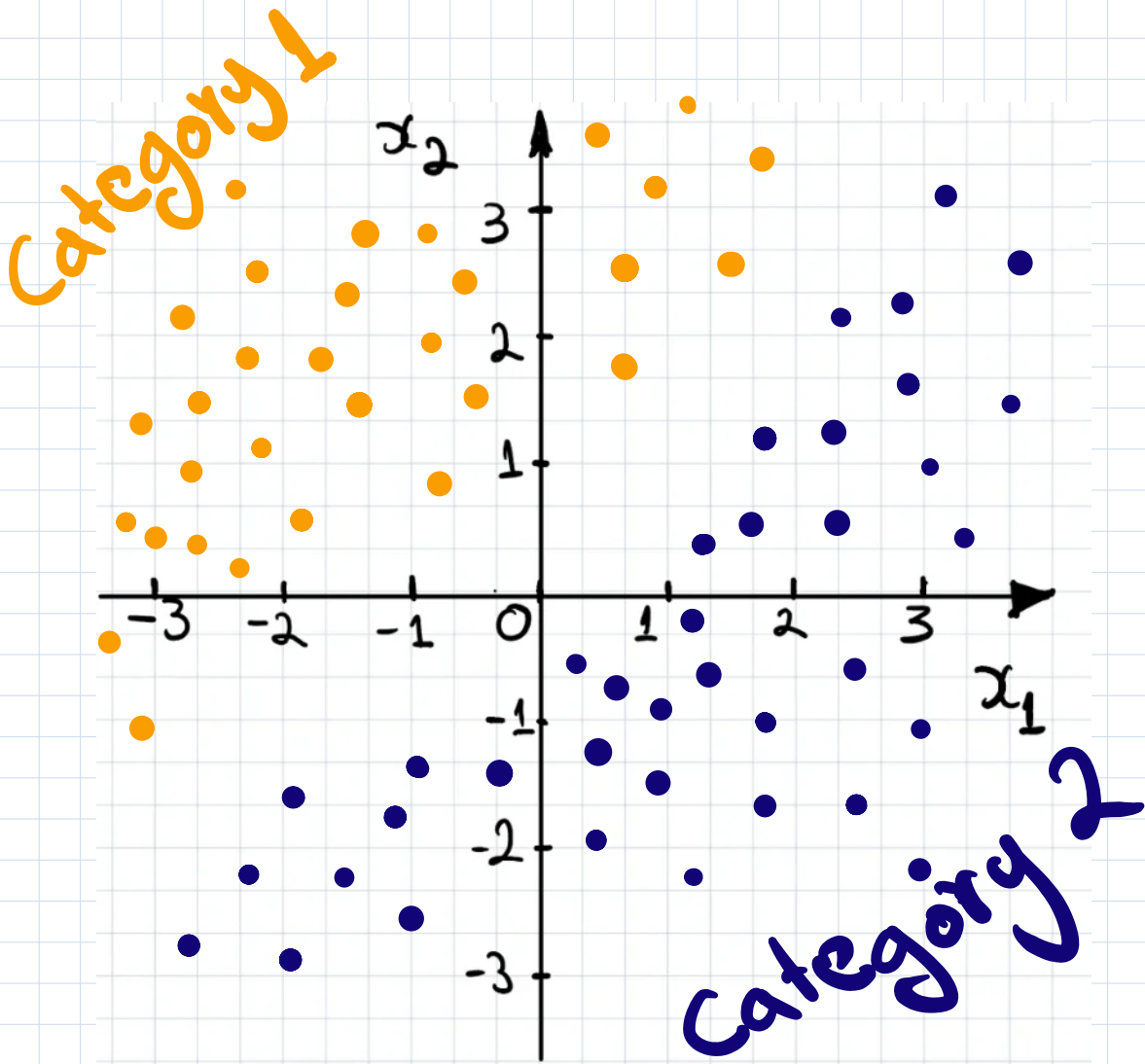


Can you sketch the
lines that can
approximate the
circle?

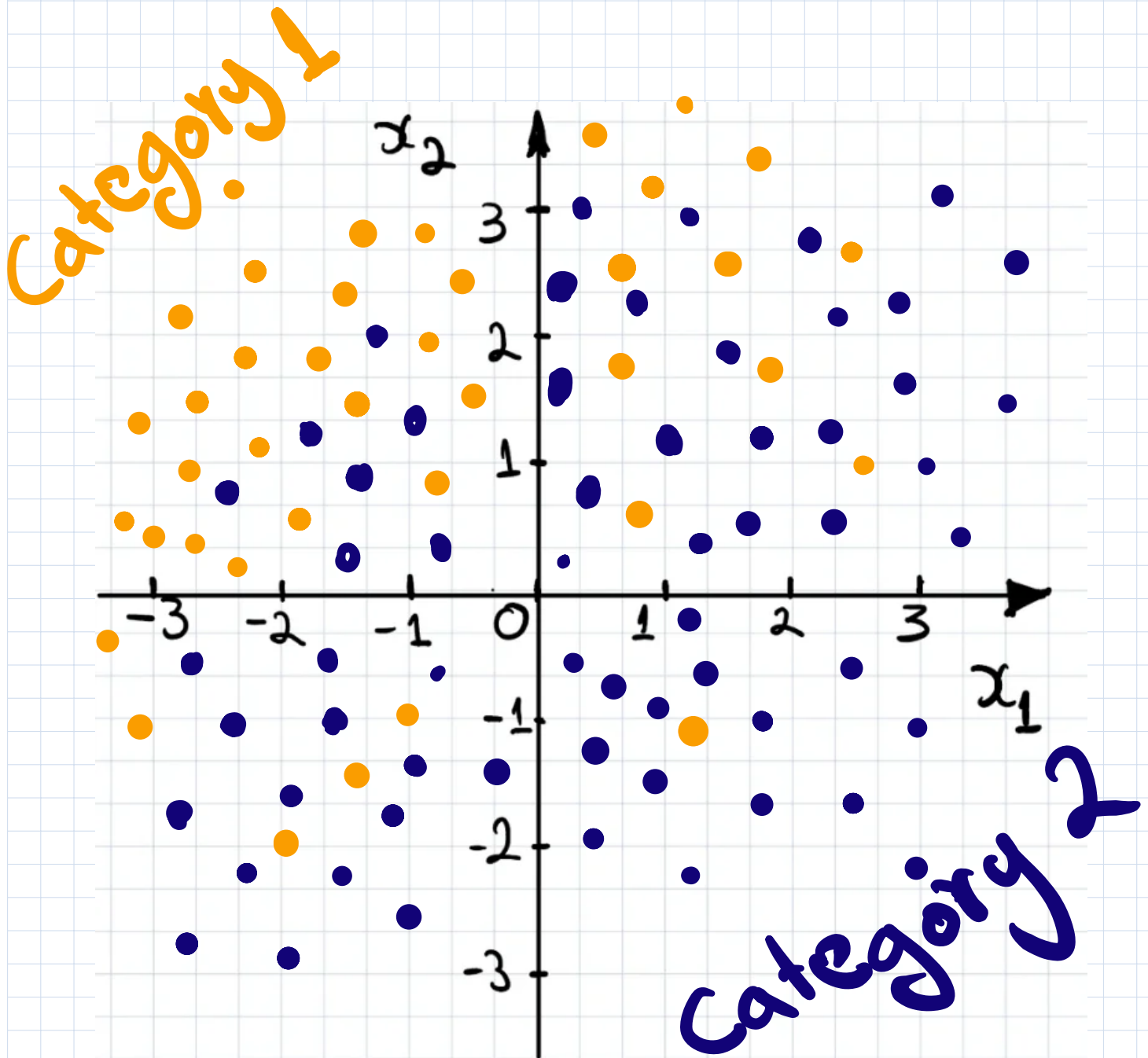


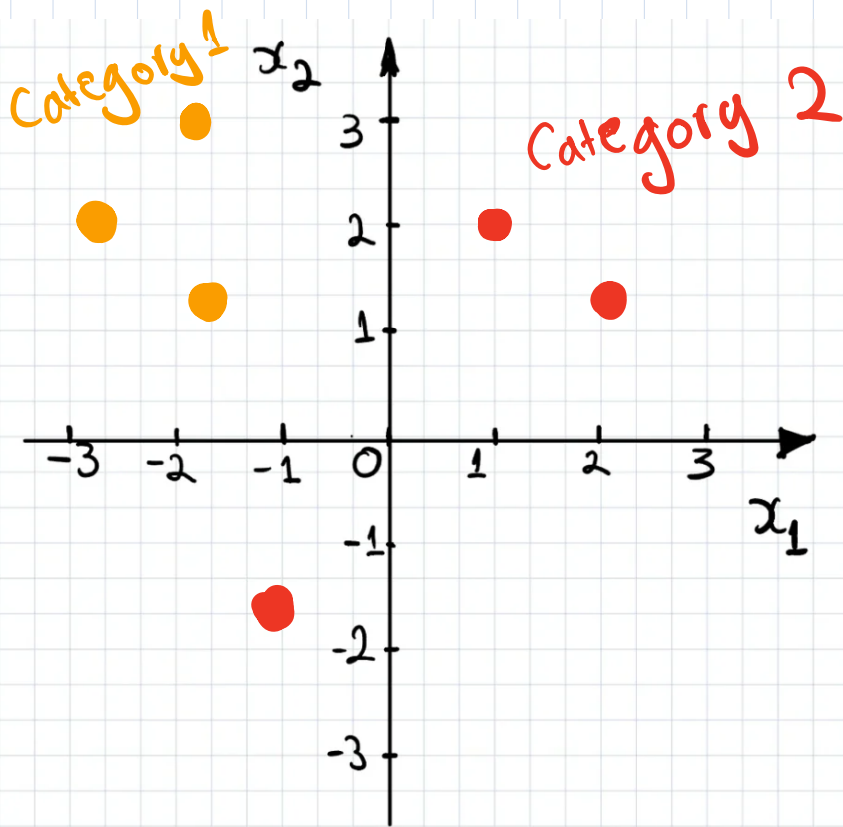
Classification:

Can you draw a line
that separates the
two colors? How?



Can we separate
them now? How?





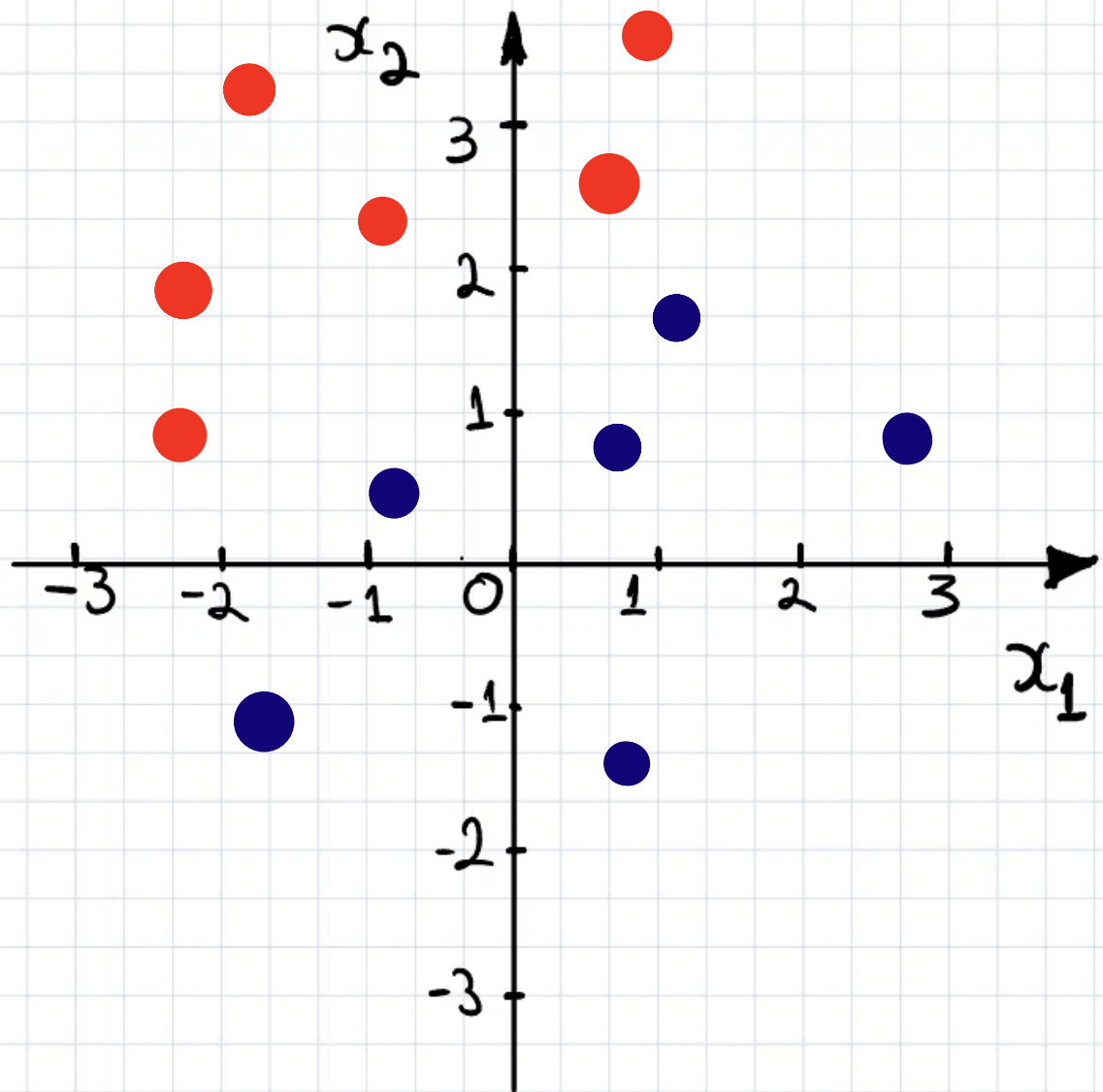
What do we do
with few
samples?

Here, we don't have
all the points!

Generalization

Goal: We want our
separation to work well
on new samples that
we have not seen.

Sampling without replacement



Sample 50% from each category (red or blue).